Problem Set 3

Starter

1 (E)**.** Using ε–δ definition of limits, prove that

(a) $\lim\limits_{n\to\infty} \dfrac{3n^2}{n^2-3} = 3.$

(b) $\lim\limits_{n\to\infty} \dfrac{3\sqrt{n}+1}{2\sqrt{n}-1} = \dfrac{3}{2}.$

2 (L-M)**.** Let $(a_n)_{n\geq 1}$ and $(b_n)_{n\geq 1}$ be two convergent sequences with $a_n \to a$ and $b_n \to b$, and let $\alpha, \beta \in \mathbb{R}$. Using the definition of limits, prove that

(a) $\lim\limits_{n\to\infty} (\alpha a_n \pm \beta b_n) = \alpha a \pm \beta b.$

(b) $\lim\limits_{n\to\infty} (a_n b_n) = ab.$

(c) If $b \neq 0$, then $\lim\limits_{n\to\infty} \dfrac{a_n}{b_n} = \dfrac{a}{b}.$

3 (L-M)**.** Prove or disprove the following statements:

(a) If $(a_n)_{n\geq 1}$ converges, then $(a_n)_{n\geq 1}$ is bounded.

(b) If $(a_n)_{n\geq 1}$ is bounded, then $(a_n)_{n\geq 1}$ converges.

(c) If $(a_n)_{n\geq 1}$ converges, then $(|a_n|)_{n\geq 1}$ converges.

(d) If $(|a_n|)_{n\geq 1}$ converges, then $(a_n)_{n\geq 1}$ converges.

(e) If $(a_n)_{n\geq 1}$ and $(b_n)_{n\geq 1}$ converge to a and b, respectively, and $a_n \leq b_n$ for all integers $n > N$ (N is a positive integer), then $a \leq b$.

(f) If $(a_n)_{n\geq 1}$ and $(b_n)_{n\geq 1}$ converge to a and b, respectively, and $a_n < b_n$ for all integers $n > N$ (N is a positive integer), then $a < b$.

4 (L-M)**.** Show that the sequence $(\sin n)_{n\geq 1}$ diverges.

Advanced

5 (L-M)**.** Let $(a_n)_{n\geq 1}$ be a real sequence.

(a) Show that if $\lim\limits_{n\to\infty} a_n = L \in \mathbb{R}$, then $\lim\limits_{n\to\infty} \dfrac{a_1 + \cdots + a_n}{n} = L.$

(b) Show that the converse of (a) is not necessarily true.

6 (M)**.** Using ε–δ definition of limits, prove that

(a) $\lim\limits_{n\to\infty} \sqrt[n]{a} = 1$, where $a > 1$

(b) $\lim\limits_{n\to\infty} \sqrt[n]{n} = 1.$

7 (M)**.** Assume that $(a_n)_{n\geq 1}$ is a sequence with $a_n \neq 0$ for which the limit $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$ exists. Show that if $L < 1$, then $\lim_{n\to\infty} a_n = 0$.

8 (H)**.** Let $(x_n)_{n\geq 1}$ and $(y_n)_{n\geq 1}$ be sequences. Show that if $(y_n)_{n\geq 1}$ is strictly increasing, $y_n \to +\infty$ and

$$\lim_{n\to\infty} \frac{x_n - x_{n-1}}{y_n - y_{n-1}} = a,$$

then

$$\lim_{n\to\infty} \frac{x_n}{y_n} = a.$$

Here a can be a real number or $\pm\infty$.